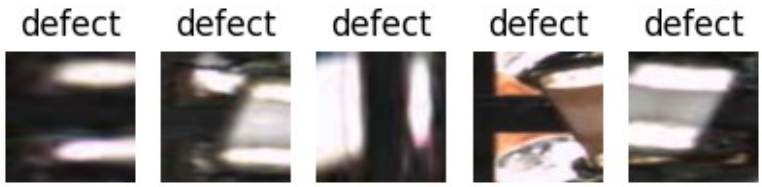
```python
import torch
import torch.nn as nn
import torch.optim as optim
import torchvision
import torchvision.transforms as transforms
from torch.utils.data import DataLoader
from torchvision import models, datasets
import matplotlib.pyplot as plt
import numpy as np
from sklearn.metrics import confusion_matrix, classification_report
import seaborn as sns

## Step 1: Load and Preprocess Data
# Define transformations for images
transform = transforms.Compose([
    transforms.Resize((224, 224)),  # Resize images for pre-trained
model input
    transforms.ToTensor(),
    #transforms.Normalize([0.485, 0.456, 0.406], [0.229, 0.224,
0.225])  # Standard normalization for pre-trained models
])

!unzip -qq ./chip_data.zip -d data

from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

```python
# Load dataset from a folder (structured as:
dataset/class_name/images)
dataset_path = "./data/dataset/"
train_dataset = datasets.ImageFolder(root=f"{dataset_path}/train",
transform=transform)
test_dataset = datasets.ImageFolder(root=f"{dataset_path}/test",
transform=transform)

# Display some input images
def show_sample_images(dataset, num_images=5):
    fig, axes = plt.subplots(1, num_images, figsize=(5, 5))
    for i in range(num_images):
        image, label = dataset[i]
        image = image.permute(1, 2, 0)  # Convert tensor format (C, H,
W) to (H, W, C)
        axes[i].imshow(image)
        axes[i].set_title(dataset.classes[label])
        axes[i].axis("off")
    plt.show()
```

```
# Show sample images from the training dataset
show_sample_images(train_dataset)
```




```
# Get the total number of samples in the training dataset
print(f"Total number of training samples: {len(train_dataset)}")

# Get the shape of the first image in the dataset
first_image, label = train_dataset[0]
print(f"Shape of the first image: {first_image.shape}")
```

```
Total number of training samples: 172
Shape of the first image: torch.Size([3, 224, 224])
```

```
# Get the total number of samples in the testing dataset
print("Number of testing samples:",len(test_dataset))
first_image1,label=test_dataset[0]


# Get the shape of the first image in the dataset

print("Image shape:",first_image1.shape)
```

```
Number of testing samples: 121
Image shape: torch.Size([3, 224, 224])
```

```
# Create DataLoader for batch processing
train_loader = DataLoader(train_dataset, batch_size=32, shuffle=True)
test_loader = DataLoader(test_dataset, batch_size=32, shuffle=False)

## Step 2: Load Pretrained Model and Modify for Transfer Learning
# Load a pre-trained VGG19 model
# write your code here
from torchvision.models import VGG19_Weights
model = models.vgg19(weights=VGG19_Weights.DEFAULT)
```

```
Downloading: "https://download.pytorch.org/models/vgg19-dcbb9e9d.pth"
to /root/.cache/torch/hub/checkpoints/vgg19-dcbb9e9d.pth

100%|██████████| 548M/548M [00:04<00:00, 123MB/s]
```

```python
# Move model to GPU if available
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

from torchsummary import summary
# Print model summary
summary(model, input_size=(3, 224, 224))
```

```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1         [-1, 64, 224, 224]           1,792
              ReLU-2         [-1, 64, 224, 224]               0
            Conv2d-3         [-1, 64, 224, 224]          36,928
              ReLU-4         [-1, 64, 224, 224]               0
         MaxPool2d-5         [-1, 64, 112, 112]               0
            Conv2d-6        [-1, 128, 112, 112]          73,856
              ReLU-7        [-1, 128, 112, 112]               0
            Conv2d-8        [-1, 128, 112, 112]         147,584
              ReLU-9        [-1, 128, 112, 112]               0
        MaxPool2d-10          [-1, 128, 56, 56]               0
           Conv2d-11          [-1, 256, 56, 56]         295,168
             ReLU-12          [-1, 256, 56, 56]               0
           Conv2d-13          [-1, 256, 56, 56]         590,080
             ReLU-14          [-1, 256, 56, 56]               0
           Conv2d-15          [-1, 256, 56, 56]         590,080
             ReLU-16          [-1, 256, 56, 56]               0
           Conv2d-17          [-1, 256, 56, 56]         590,080
             ReLU-18          [-1, 256, 56, 56]               0
        MaxPool2d-19          [-1, 256, 28, 28]               0
           Conv2d-20          [-1, 512, 28, 28]       1,180,160
             ReLU-21          [-1, 512, 28, 28]               0
           Conv2d-22          [-1, 512, 28, 28]       2,359,808
             ReLU-23          [-1, 512, 28, 28]               0
           Conv2d-24          [-1, 512, 28, 28]       2,359,808
             ReLU-25          [-1, 512, 28, 28]               0
           Conv2d-26          [-1, 512, 28, 28]       2,359,808
             ReLU-27          [-1, 512, 28, 28]               0
        MaxPool2d-28          [-1, 512, 14, 14]               0
           Conv2d-29          [-1, 512, 14, 14]       2,359,808
             ReLU-30          [-1, 512, 14, 14]               0
           Conv2d-31          [-1, 512, 14, 14]       2,359,808
             ReLU-32          [-1, 512, 14, 14]               0
           Conv2d-33          [-1, 512, 14, 14]       2,359,808
             ReLU-34          [-1, 512, 14, 14]               0
           Conv2d-35          [-1, 512, 14, 14]       2,359,808
             ReLU-36          [-1, 512, 14, 14]               0
        MaxPool2d-37            [-1, 512, 7, 7]               0
AdaptiveAvgPool2d-38            [-1, 512, 7, 7]               0
           Linear-39                 [-1, 4096]     102,764,544
```

```
             ReLU-40                  [-1, 4096]               0
          Dropout-41                  [-1, 4096]               0
           Linear-42                  [-1, 4096]      16,781,312
             ReLU-43                  [-1, 4096]               0
          Dropout-44                  [-1, 4096]               0
           Linear-45                  [-1, 1000]       4,097,000
================================================================
Total params: 143,667,240
Trainable params: 143,667,240
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 238.69
Params size (MB): 548.05
Estimated Total Size (MB): 787.31
----------------------------------------------------------------
```

```python
# Modify the final fully connected layer to match the dataset classes
# Write your code here
model.classifier[-1]=nn.Linear(model.classifier[-1].in_features,1)


# Move model to GPU if available
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

summary(model, input_size=(3, 224, 224))
```

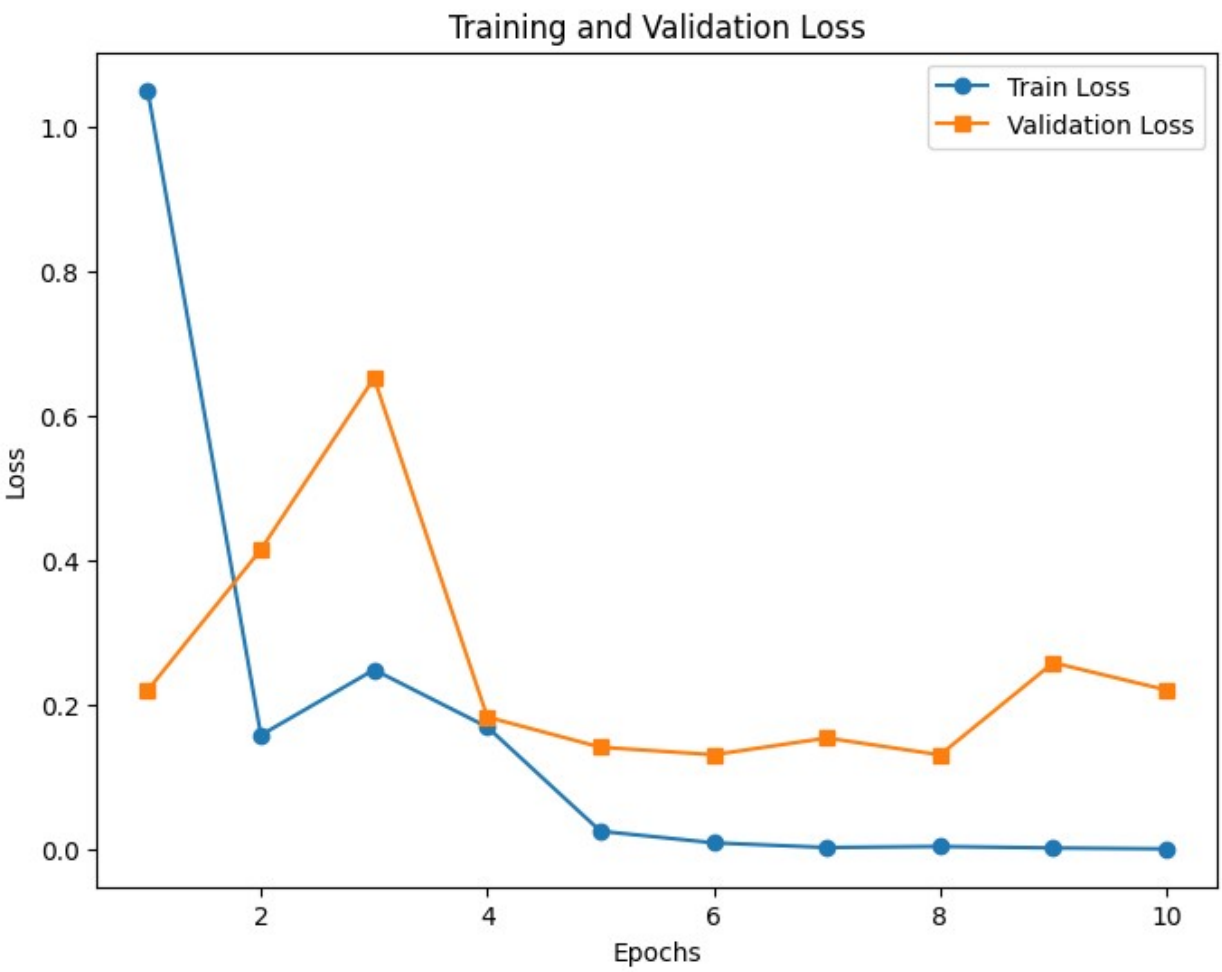
```
----------------------------------------------------------------
        Layer (type)               Output Shape         Param #
================================================================
            Conv2d-1         [-1, 64, 224, 224]           1,792
              ReLU-2         [-1, 64, 224, 224]               0
            Conv2d-3         [-1, 64, 224, 224]          36,928
              ReLU-4         [-1, 64, 224, 224]               0
         MaxPool2d-5         [-1, 64, 112, 112]               0
            Conv2d-6        [-1, 128, 112, 112]          73,856
              ReLU-7        [-1, 128, 112, 112]               0
            Conv2d-8        [-1, 128, 112, 112]         147,584
              ReLU-9        [-1, 128, 112, 112]               0
        MaxPool2d-10          [-1, 128, 56, 56]               0
           Conv2d-11          [-1, 256, 56, 56]         295,168
             ReLU-12          [-1, 256, 56, 56]               0
           Conv2d-13          [-1, 256, 56, 56]         590,080
             ReLU-14          [-1, 256, 56, 56]               0
           Conv2d-15          [-1, 256, 56, 56]         590,080
             ReLU-16          [-1, 256, 56, 56]               0
           Conv2d-17          [-1, 256, 56, 56]         590,080
             ReLU-18          [-1, 256, 56, 56]               0
        MaxPool2d-19          [-1, 256, 28, 28]               0
```

```
            Conv2d-20          [-1, 512, 28, 28]       1,180,160
              ReLU-21          [-1, 512, 28, 28]               0
            Conv2d-22          [-1, 512, 28, 28]       2,359,808
              ReLU-23          [-1, 512, 28, 28]               0
            Conv2d-24          [-1, 512, 28, 28]       2,359,808
              ReLU-25          [-1, 512, 28, 28]               0
            Conv2d-26          [-1, 512, 28, 28]       2,359,808
              ReLU-27          [-1, 512, 28, 28]               0
         MaxPool2d-28          [-1, 512, 14, 14]               0
            Conv2d-29          [-1, 512, 14, 14]       2,359,808
              ReLU-30          [-1, 512, 14, 14]               0
            Conv2d-31          [-1, 512, 14, 14]       2,359,808
              ReLU-32          [-1, 512, 14, 14]               0
            Conv2d-33          [-1, 512, 14, 14]       2,359,808
              ReLU-34          [-1, 512, 14, 14]               0
            Conv2d-35          [-1, 512, 14, 14]       2,359,808
              ReLU-36          [-1, 512, 14, 14]               0
         MaxPool2d-37            [-1, 512, 7, 7]               0
AdaptiveAvgPool2d-38            [-1, 512, 7, 7]               0
            Linear-39                 [-1, 4096]     102,764,544
              ReLU-40                 [-1, 4096]               0
           Dropout-41                 [-1, 4096]               0
            Linear-42                 [-1, 4096]      16,781,312
              ReLU-43                 [-1, 4096]               0
           Dropout-44                 [-1, 4096]               0
            Linear-45                    [-1, 1]           4,097
================================================================
Total params: 139,574,337
Trainable params: 139,574,337
Non-trainable params: 0
----------------------------------------------------------------
Input size (MB): 0.57
Forward/backward pass size (MB): 238.68
Params size (MB): 532.43
Estimated Total Size (MB): 771.69
----------------------------------------------------------------
```

```python
# Freeze all layers except the final layer
for param in model.features.parameters():
    param.requires_grad = False  # Freeze feature extractor layers

# Include the Loss function and optimizer
criterion =nn.BCEWithLogitsLoss()
optimizer =optim.Adam(model.parameters(), lr=0.001)

## Step 3: Train the Model
def train_model(model, train_loader,test_loader,num_epochs=100):
    train_losses=[]
    val_losses=[]
    model.train()
```

```
    for epoch in range(num_epochs):
        running_loss=0.0
        for images,labels in train_loader:
            images,labels=images.to(device),labels.to(device)
            optimizer.zero_grad()
            outputs=model(images)
            loss=criterion(outputs,labels.unsqueeze(1).float())

            loss.backward()
            optimizer.step()
            running_loss+=loss.item()
        train_losses.append(running_loss/len(train_loader))

        # Compute validation loss
        model.eval()
        val_loss=0.0
        with torch.no_grad():
          for images,labels in test_loader:
            images,labels=images.to(device),labels.to(device)
            outputs=model(images)
            loss=criterion(outputs,labels.unsqueeze(1).float())
            val_loss+=loss.item()
        val_losses.append(val_loss/len(test_loader))
        model.train()
        print(f'Epoch [{epoch+1}/{num_epochs}], Train Loss:
{train_losses[-1]:.4f}, Validation Loss: {val_losses[-1]:.4f}')
    # Plot training and validation loss
    print("Name: Y.NITHEESH ")
    print("Register Number: 212224040370")
    plt.figure(figsize=(8, 6))
    plt.plot(range(1, num_epochs + 1), train_losses, label='Train
Loss', marker='o')
    plt.plot(range(1, num_epochs + 1), val_losses, label='Validation
Loss', marker='s')
    plt.xlabel('Epochs')
    plt.ylabel('Loss')
    plt.title('Training and Validation Loss')
    plt.legend()
    plt.show()

# Move model to GPU if available
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model = model.to(device)

# Train the model
# Write your code here
train_model(model,train_loader,test_loader,num_epochs=10)
```

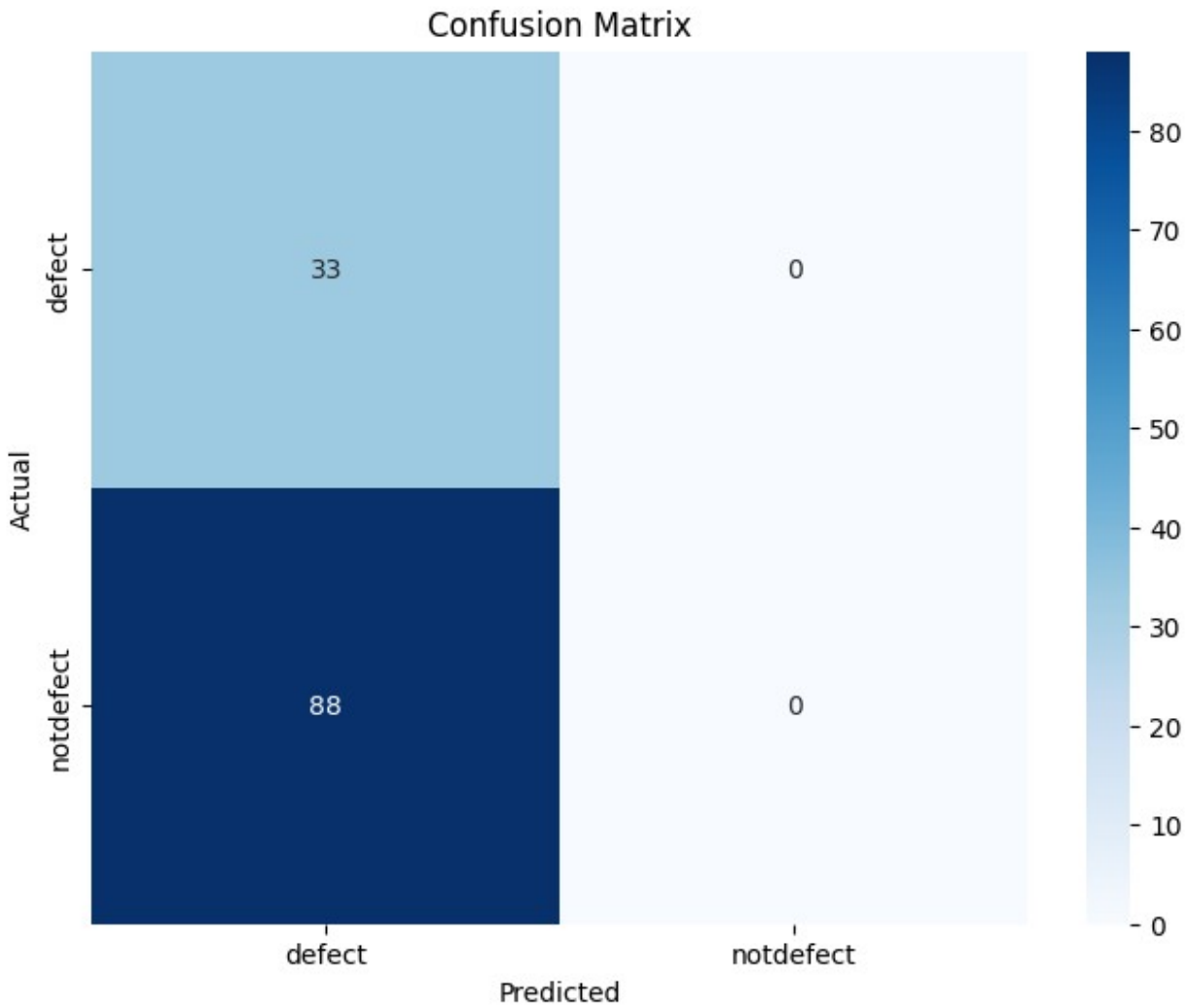
```
Epoch [1/10], Train Loss: 1.0514, Validation Loss: 0.2197
Epoch [2/10], Train Loss: 0.1577, Validation Loss: 0.4148
Epoch [3/10], Train Loss: 0.2482, Validation Loss: 0.6525
Epoch [4/10], Train Loss: 0.1688, Validation Loss: 0.1827
Epoch [5/10], Train Loss: 0.0244, Validation Loss: 0.1409
Epoch [6/10], Train Loss: 0.0084, Validation Loss: 0.1304
Epoch [7/10], Train Loss: 0.0018, Validation Loss: 0.1536
Epoch [8/10], Train Loss: 0.0033, Validation Loss: 0.1303
Epoch [9/10], Train Loss: 0.0012, Validation Loss: 0.2580
Epoch [10/10], Train Loss: 0.0000, Validation Loss: 0.2201
Name: Y.NITHEESH
Register Number: 212224040370
```




```python
## Step 4: Test the Model and Compute Confusion Matrix &
Classification Report
def test_model(model, test_loader):
    model.eval()
    correct = 0
    total = 0
```

```
    all_preds = []
    all_labels = []

    with torch.no_grad():
        for images, labels in test_loader:
            images, labels = images.to(device), labels.to(device)
            outputs = model(images)
            _, predicted = torch.max(outputs, 1)
            total += labels.size(0)
            correct += (predicted == labels).sum().item()
            all_preds.extend(predicted.cpu().numpy())
            all_labels.extend(labels.cpu().numpy())

    accuracy = correct / total
    print(f'Test Accuracy: {accuracy:.4f}')

    # Compute confusion matrix
    cm = confusion_matrix(all_labels, all_preds)
    print("Name: Y.NITHEESH       ")
    print("Register Number: 212224040370")
    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
xticklabels=train_dataset.classes, yticklabels=train_dataset.classes)
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.title('Confusion Matrix')
    plt.show()

    # Print classification report
    print("Name: Y.NITHEESH     ")
    print("Register Number: 212224040370       ")
    print("Classification Report:")
    print(classification_report(all_labels, all_preds,
target_names=train_dataset.classes))

# Evaluate the model
# write your code here
test_model(model,test_loader)
```

```
Test Accuracy: 0.2727
Name: Y.NITHEESH
Register Number: 212224040370
```

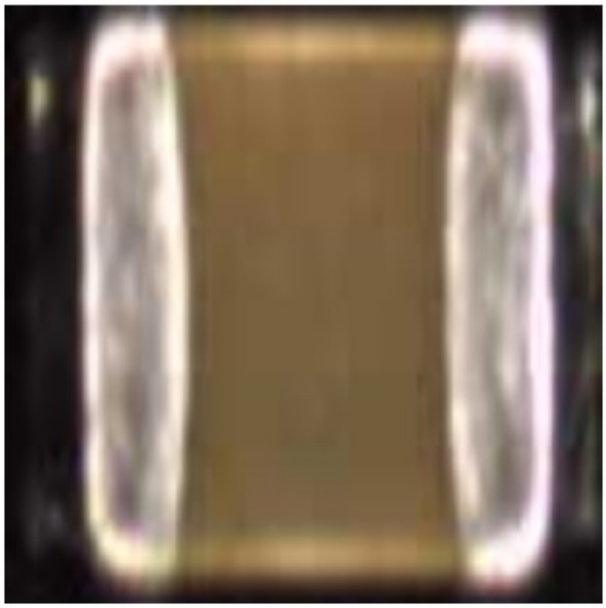



```
Name: Y.NITHEESH
Register Number: 212224040370
Classification Report:
              precision    recall  f1-score   support

      defect       0.27      1.00      0.43        33
   notdefect       0.00      0.00      0.00        88

    accuracy                           0.27       121
   macro avg       0.14      0.50      0.21       121
weighted avg       0.07      0.27      0.12       121


/usr/local/lib/python3.12/dist-packages/sklearn/metrics/
_classification.py:1565: UndefinedMetricWarning: Precision is ill-
defined and being set to 0.0 in labels with no predicted samples. Use
`zero_division` parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
```

```
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classificatio
n.py:1565: UndefinedMetricWarning: Precision is ill-defined and being
set to 0.0 in labels with no predicted samples. Use `zero_division`
parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
/usr/local/lib/python3.12/dist-packages/sklearn/metrics/_classificatio
n.py:1565: UndefinedMetricWarning: Precision is ill-defined and being
set to 0.0 in labels with no predicted samples. Use `zero_division`
parameter to control this behavior.
  _warn_prf(average, modifier, f"{metric.capitalize()} is",
len(result))
```

```python
## Step 5: Predict on a Single Image and Display It
def predict_image(model, image_index, dataset):
    model.eval()
    image, label = dataset[image_index]
    with torch.no_grad():
        image_tensor = image.unsqueeze(0).to(device)
        output = model(image_tensor)

        # Apply sigmoid to get probability, threshold at 0.5
        prob = torch.sigmoid(output)
        predicted = (prob > 0.5).int().item()


    class_names = class_names = dataset.classes
    # Display the image
    image_to_display = transforms.ToPILImage()(image)
    plt.figure(figsize=(4, 4))
    plt.imshow(image_to_display)
    plt.title(f'Actual: {class_names[label]}\nPredicted:
{class_names[predicted]}')
    plt.axis("off")
    plt.show()

    print(f'Actual: {class_names[label]}, Predicted:
{class_names[predicted]}')

# Example Prediction
predict_image(model, image_index=55, dataset=test_dataset)
```

Actual: notdefect
Predicted: notdefect



```
Actual: notdefect, Predicted: notdefect

#Example Prediction
predict_image(model, image_index=25, dataset=test_dataset)
```

Actual: defect
Predicted: defect

```
Actual: defect, Predicted: defect
```